

Welcome to **instats**

The session will begin shortly.

Start

A 3-DAY WORKSHOP ON LLMS FOR MEASUREMENT

What is an LLM, and which kind do I want?

An intuition-first tour through encoder and generative models, and the small set of decisions that separates them.

INSTRUCTOR

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AUDIENCE

Political scientists, no Python required

DURATION

2 hours 30 minutes

Three movements, plus your first Colab.

MOVEMENT 01, 35 MIN

The model zoo

Encoders read sentences and return labels. Generative models continue text. Why the distinction shapes your research design.

MOVEMENT 02, 30 MIN

The Colab tour

What Google Colab is, why it has a GPU, and how it lets you run a state-of-the-art encoder from your browser.

MOVEMENT 03, 35 MIN

Stance on real UN speeches

We run a real zero-shot encoder on roughly two hundred sentences from the UN General Debate Corpus and see what it gets right and where it stumbles.

Closing 15 minutes are hands-on. You open free Colab on your own Google account and run a tiny version yourself.

Three published examples, three different LLM jobs.

JOB 01, STANCE

Bestvater & Monroe (2023)

Sentiment Is Not Stance. Political Analysis 31(2).

Off-the-shelf sentiment scores conflate affect with stance toward a target. A target-aware fine-tuned encoder outperforms generic sentiment.

Encoder. Supervised, fine-tuned on hand-codes.

ENCODER

JOB 02, TOPIC

Wankmüller (2024)

Neural transfer learning for political text. Political Analysis.

Fine-tuned BERT-class encoders scale issue-topic classification across long parliamentary corpora that LDA-era methods could not reach.

Encoder. Fine-tuned on hundreds to thousands of hand-codes.

ENCODER

JOB 03, EXTRACTION

Ornstein et al. (2025)

How to Train Your Stochastic Parrot. PSRM 13(2).

Few-shot prompts to a generative LLM often match conventional supervised pipelines on text-as-data tasks, with caveats around cost and reproducibility.

Generative.

GENERATIVE

Same field, same recent year span, two model families. Today we learn how to choose between them.

How to read today.

RULE 01

Intuition first

No matrix algebra. No transformer block diagrams. You will leave today able to make principled choices about model families without internalizing equations.

RULE 02

Demos are mostly mine

Instructor demos run on paid Colab Pro for speed. The hands-on at the end is the one segment you run yourself, on free Colab.

RULE 03

We choose carefully

The most important skill today is knowing when an encoder is the right tool and when a generative model is. For measurement work, the answer is usually the encoder. We will see why.

BEFORE WE BEGIN



The bottleneck for political science measurement is no longer compute or model availability. It is the validity of the labels we ask the model to produce.

Three quick checks while we settle in. One. Are you signed into a Google account in this browser? Two. Click the hands-on link in chat and confirm it opens in Colab. You do not need to run anything yet. Three. Type "ready" in chat once both checks pass.

WORKSHOP FRAMING, DAY 1

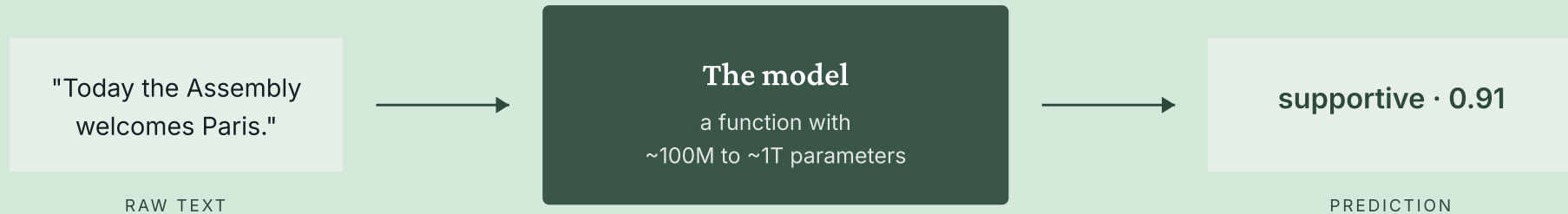
FIRST MOVEMENT, 35 MINUTES

The plain-language model zoo.

Two families of models. Five questions to decide which one your research question wants.

01

An LLM turns text into numbers, then numbers into predictions.



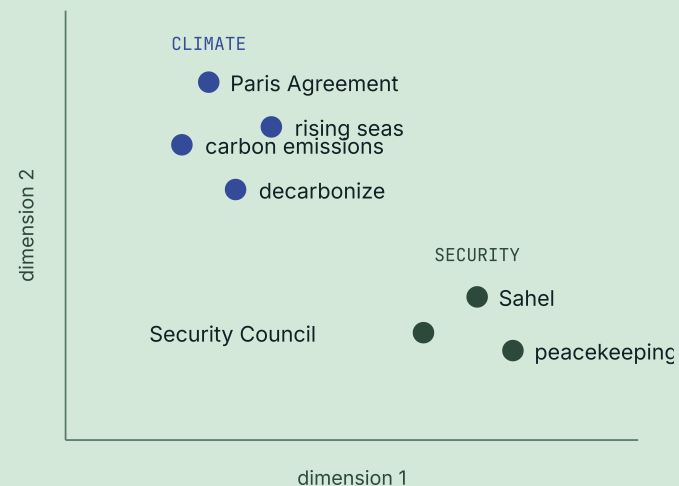
The interesting question is what kind of prediction comes out.

Words live in a high-dimensional space. Similar meanings sit close together.

Modern models do not store words as letters. Each token is stored as a list of about a thousand numbers. That list is a position in a "meaning space" learned from billions of training sentences.

The geometry is what makes everything else possible. Once "Paris Agreement" and "carbon emissions" sit near each other, every downstream task that uses those numbers gets a head start.

Think of it as a structured replacement for bag-of-words and TF-IDF, where the structure is meaning rather than co-occurrence.



Same starting geometry. Two very different jobs.

FAMILY 01

Encoders

Read a piece of text in full and output a fixed prediction. A label, a score, a probability.

BERT · RoBERTa · DeBERTa · ModernBERT

Smaller, faster, deterministic. Run on a laptop or free Colab CPU.

DAY 1 + DAY 2

FAMILY 02

Generative models

Continue text one token at a time. Given a prompt, produce more text. The prompt steers the output.

GPT · Claude · Gemini · LLaMA · Qwen

Larger, slower, stochastic. Usually called via API.

DAY 3

Both descend from the same Transformer architecture (2017). They diverge in what they were trained to predict. That difference shapes everything downstream.

Read the whole sentence. Return a number.

WHAT KINDS OF NUMBERS

- Stance toward a target
- Sentiment, tone, emotion
- Topic and framing labels
- Named entity tags
- Anything where you need a fixed, comparable score per text

WHY POLITICAL SCIENTISTS LOVE THEM

- **Deterministic.** Same input, same output. Run the same model in 2026 and 2030 and you get the same answer.
- **Cheap.** After the model downloads once, inference is essentially free.
- **Validatable.** The whole machinery of cross-validation and confusion matrices is right there.

For measurement projects, the default is encoder.

Continue the text, token by token.

USE IT FOR

- Structured extraction from messy text
- Free-text summarization
- Normalization of inconsistent fields
- Reasoning over long context
- Anything where the answer cannot be a label from a finite list

WHAT YOU GIVE UP

- **Stochastic by default.** Same prompt, different answers. Pin temperature 0.
- **Higher cost per call.** Tokens add up.
- **Reproducibility tax.** Pin model versions. Log every prompt and response.

Reach for it when free-form output is the only fit. Day 3 is the deep dive.

The slide you screenshot.

	CLOSED LABEL SET	FREE-FORM OUTPUT	MIXED
Small data ~hundreds to thousands	ENCODER, ZERO-SHOT Off-the-shelf DeBERTa with your own labels. No training.	GENERATIVE, FEW-SHOT Claude or Gemini with 3 to 5 examples in the prompt.	GENERATIVE, PROMPTED LLM with clear schema and validators.
Large data ~tens of thousands and more	ENCODER, FINE-TUNED 300 hand-labels and you can train an encoder.	ENCODER + GENERATIVE Encoder filter, generative for hard cases.	HYBRID PIPELINE Generative for extraction, encoder for classification.

If the task can be written as "give me a fixed label," start with an encoder. Reach for generative when the answer cannot fit in one.

The NLI trick.

One skill, every classification task.

An NLI-trained encoder learns one job. Given a **premise** and a **hypothesis**, decide whether the premise entails the hypothesis, contradicts it, or is neutral.

NLI stands for natural language inference. The DeBERTa we use today was pretrained on three NLI corpora that together contain millions of premise-hypothesis pairs.

That single skill is shockingly general. Any classification task can be rewritten as a yes-or-no entailment question with a clever choice of hypothesis.

EXAMPLES

- **Stance.** "Does this passage support stronger climate action?"
- **Topic.** "Is this paragraph about agricultural trade?"
- **Framing.** "Does this speech frame migration as a security issue?"

```
zsc(text, candidate_labels=[
    "supports climate action",
    "skeptical of climate action",
    "does not discuss climate",
])
# returns one probability per label
```

The same machinery, scaled to a published proxy.

One quick example from my R&R paper at the *Review of International Organizations*.

Each paragraph from a US Trade Representative report is the premise. Thirteen claims about IP enforcement are the hypotheses. A weighted sum of the entailment probabilities becomes a country-year proxy for US assessments of foreign IP regimes, 1995 to 2022.

The same off-the-shelf NLI machinery you will run on UN speeches in fifteen minutes scales to a published measurement design with the right hypothesis engineering.

Only Day 1 slide that mentions my own work. Everything else today is off-the-shelf on public data.



Five things you now know.

- The split between encoder and generative.
- The default for measurement projects, which is encoder.
- The escape hatch, which is generative.
- The NLI trick.
- The reproducibility tax that comes with generative.

Take a ten-minute break. We come back at the time on screen and look at what Google Colab actually is.

Stop